INCH POUND
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

LEAK DETECTOR, FUEL SYSTEM

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL -L- 83774 for all federal agencies.

1. SCOPE.

1.1 General. This Commercial Item Description (CID) describes an explosion-proof leak detector for locating fuel leaks in aircraft fuel cells and tanks using helium, argon, or nitrous oxide as a tracer media.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: (SA-ALC/TILDD, 485 Quentin Roosevelt Rd, Kelly AFB TX 78241) by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4940

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2. SALIENT CHARACTERISTICS.

2.1 Design. The leak detector shall weigh no more than 25 pounds and shall be operable by one man. The power supply shall be capable of operating the sensing unit a minimum of 4 hours without recharging. The leak detector will be explosion proof for operation in an invironment defined by the National Electrical Code, NFPA 70, as class I, division I, group D. The leak detector will be safe for use in an explosive atmosphere at 8000 feet altitude and 125 degrees Fahrenheit. The leak detector may be enclosed in one or two housings. The weight of the hand held unit for leak searching shall not exceed 8 pounds. The dimensions of the housing shall not exceed 6 inches wide by 12 inches long by 10 inches deep. If two seperate housings are utilized, the connecting cord shall be 8 feet in length with provisions for extending the working distance an additional 8 feet. The housings shall incorporate one common static electricity grounding point. A retractable (coiled) cable and spring clip shall be attached to that ground point for grounding to the aircraft. The cable retracted length shall be 12 inches. The extended length shall be 6 feet. The leak detector shall be furnished complete with all items and accessories required for proper operation. A meter as well as audio and light indicators shall be the means by which leaks are indicated. The sampling probe shall be flexible, semi-rigid, non-kinking, with a minimum length of 12 inches. A Carrying Case shall be supplied with the leak detector to provide for storage and ease of transportation.

2.2 Component Characteristics.

2.2.1 Sensing unit. The sensing unit shall be capable of detecting the following gasses at the specified leak rate when tested in accordance with paragraph 4.4.2:

Helium	1×10^{-3} cubic centimeter per second
Argon	4×10^{-3} cubic centimeter per second
Nitrous Oxide	4×10^{-3} cubic centimeter per second

2.2.2 Response time. When utilizing the sampling probe, the sensing unit full scale deflection response time to a 1×10^{-3} cubic centimeter per second helium leak shall not exceed one second.

2.2.3 Recovery time. When removed from a leak source, the sensing unit recovery time (meter return to 10 percent of full scale) shall not exceed 3 seconds.

2.2.4 Interference due to background gases. The sensing unit shall be capable of detecting a 5 percent by volume concentration of any of the tracer gases when operating in an atmosphere containing a minimum of 0.4 percent by volume concentration of JP-8 fuel vapor.

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2.2.5 Range capability. The sensing unit shall be capable of locating a leak of 1 cubic centimeter per second helium and a leak of 1 X 10^{-3} cubic centimeters per second helium when the leaks are 4 inches apart horizontally. There shall be evidence that two seperate leaks have been located. This shall be accomplished in still air without utilizing range swiches, zero adjustment, or other controls.

2.2.6 Power supply. The power supply shall be an internal power source capable of operating the leak detector for periods of 4 hours continuously and shall be rechargeable. Batteries shall be capable of a minimum of 500 recharging cycles from the voltage cut-off depth of discharge to a fully charged condition. The power supply shall be capable of preventing erroneous readings or indications due to low voltage. The power supply shall include an internal recharger capable of recharging the batteries from a 110-120 volt AC, 50-60 cycle outlet. A detachable 3-wire charging cord (minimum length of 6 feet) shall be included. These requirements shall be obtained when the power supply is tested in accrdance with paragraph 4.4.7.

3. REGULATORY REQUIREMENTS.

3.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS.

4.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producers own drawings, specifications, standards, and quality assurance practices, and be the same product (explosion proof) offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Inspection and testing. The inspection and testiing of the leak detector shall be performed as indicated in this section.

4.3 Test conditions. Unless otherwise specified, all tests required by this Commercial Item Description (CID) shall be made at an atmosphereic pressure of 28 to 32 inches of mercury, a temperature of 25 ± 5^{0} centigrade (C), and a relative humidity of 80 percent or less.

4.4 Test methods. All tests involving any aspect of leak detection shall be performed with the probe tip no closer than 3/8 inch to the leak.

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4.4.1 Noise level and zero drift test. The leak detector shall be checked to determine that the inherent electrical noise causes no more than 5 percent meter displacement. During the tests referenced in paragraphs 4.4.2, 4.4.3, and 4.4.4, the leak detector shall be checked to determine that the zero drift is not more than 5 percent meter displacement. This check shall be for a minimum of 30 minutes.

4.4.2 Sensitivity test. The leak detector shall be exposed to a 1 X 10^{-3} cubic centimeters per second helium leak. A full scale deflection on the meter shall be recorded. This test shall be repeated two times. Any reading less than full scale either time shall be cause for rejection. The light and audio indicator shall be checked for visibility, audibility, and initiation at 30 percent ±5 percent full scale meter deflection. This same procedure shall be repeated for argon and nitrous oxide leaks of 4 X 10^{-3} cubic centimeters per second.

4.4.3 Response time test. The leak detector shall be tested for compliance with paragraph 2.2.2.

4.4.4 Recovery time test. The leak detector shall be tested for compliance with paragraph 2.2.3.

4.4.5 Interference test. The leak detector shall be tested for compliance with paragraph 2.2.4.

4.4.6 Range test. The leak detector shall be tested for compliance with paragraph 2.2.5.

4.4.7 Power supply test. The leak detector shall be operated for 4 hours. During this 4 hour period the power supply shall be positioned 1 hour on each side and 1 hour on each end. The leak detector shall be exposed to a 1 X 10^{-3} cubic centimeters per second helium leak every 10 minutes. Sensitivity, response time, light, and audio indication shall be tested for compliance with paragraphs 4.4.2, 4.4.3, and 4.4.4. Any evidence of malfunctioning shall be cause for rejection. The leak detector shall then be operated until the voltage cutoff or other automatic device is actuated. The recharger shall be activated in acordance with manufacturers instructions allowing the batteries to be fully recharged in no more than 14 hours.

4.4.8 Explosion-proof test. Either Underwriters Laboratories, Inc certification as explosion-proof or factory mutual research approval as intrinsically safe for operation in a class I, division 1, group D environment is acceptable for meeting the explosion-proof testing requirement.

5. PACKAGING.

5.1 Preservation, packing, and marking shall be as specified in the contract or order.

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6. NOTES

6.1 Source of Document.

6.1.1 Copies of the National Electrical Code may be obtained from the National Fire Protection Association, Batterymarch Park, Quincy MA 02269.

6.2 National Stock Numbers (NSNs). The following is a list of NSNs assigned which correspond to this CID. This list may not be indicative of all posible NSNs associated with the CID.

NSN
4940-01-232-0694

MILITARY INTERESTS:
Custodians:
Air Force - 99

Preparing Activity:
Air Force - 82
Agent:
Air Force - 99

CIVIL AGENCY
COORDINATING ACTIVITIES:
Reviewing Activity:
GSA-FSS

Project Number 4940-0678

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